Vision-Language Models for Spatial Understanding

Diffusion Generation, Proxy Evaluation, and VLM-as-a-Judge

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Motivation

Problem

- Text-to-image diffusion models generate realistic images, but often fail at **compositional understanding**.

Typical failures

- wrong 2D spatial relations
- wrong 3D relations
- wrong object counts
- failures in multi-object scenes

Examples

- left / right / top / bottom
- in front of / behind / hidden by
- exact counting

Project Goal

Main goal

- Evaluate compositional consistency in generated images
- Study whether **VLMs** can work as judges of prompt-image alignment

Main components

- Compare two generators:
  - **SDXL**
  - **FLUX**
- Build an automatic **proxy pipeline**
- Compare proxy outputs against **VLM-as-a-Judge** outputs

Experimental Setup

Benchmark categories

- spatial
- 3d_spatial
- numeracy
- complex

Generators

- SDXL
- FLUX
- 3 seeds per prompt

VLM judges

- Qwen2.5-VL
- Qwen3-VL
- Gemma 3

Prompt variants

- prompt_v1
- prompt_v2

Notebook 1: Image Generation

What I did

- Loaded prompts from the four benchmark categories
- Generated images with:
  - SDXL
  - FLUX
- Used **3 seeds per prompt**
- Saved images and indexing CSV files

Why

- Build a comparable image dataset for downstream evaluation
- Reduce dependence on one stochastic output

Notebook 2: Automatic Proxy Construction

What I did

- Built an automatic proxy pipeline
- Step 2:
  - detect prompt objects with OWL-ViT
  - infer coarse relations from boxes
- Step 3:
  - refine detections with SAM3
  - compute mask-based spatial reasoning

Also included

- presence proxy
- relation proxy
- numeracy proxy

Why

- Generated images do not have human labels
- Proxy provides an automatic reference baseline

Notebook 3: VLM-as-a-Judge

What I did

- Evaluated images with:
  - Qwen2.5-VL
  - Qwen3-VL
  - Gemma 3
- Tested:
  - prompt_v1
  - prompt_v2

Judge input

- generated image
- original prompt

Judge output

- presence score (1–5)
- relation / count / composition score (1–5)

Why

- Add a semantic evaluator independent from the proxy

Notebook 4: Final Demo and Analysis

Demo path

- small subset selection
- regeneration
- Step 2
- Step 3
- numeracy
- one VLM judge

Analysis path

- load full saved outputs
- compare generators, categories, judges, and prompts
- interpret the full experiment results

Evaluation Metrics

Score-based metrics

- mean proxy scores
- mean VLM scores
- mean **normalized VLM scores**
- Pearson correlation
- Spearman correlation

Agreement / robustness

- Cohen's kappa
- weighted kappa
- binary accuracy
- JSON parse rate

Normalization

- VLM scores were also rescaled from 1–5 to 0–1 to make comparison with proxy scores easier

Proxy Results by Category

Split	FLUX	SDXL
3d_spatial	0.352	0.357
complex	0.880	0.857
numeracy	0.277	0.269
spatial	0.618	0.597

Main finding

- Difficulty ranking:
    - **complex** easiest
    - **spatial** next
    - **3d_spatial** harder
    - **numeracy** hardest

Generator Comparison

Overall proxy relation mean

$$\text{FLUX } 0.532 > \text{SDXL } 0.520$$

Interpretation

- FLUX is slightly stronger overall under the proxy baseline
- The difference is not huge, but it is consistent in the overall average
- Generator ranking still depends on the metric:
  - proxy averages
  - correlation with VLM
  - agreement metrics

VLM Judge Comparison

Judge	Pearson rel.	Strict kappa rel.	Parse rate
Gemma 3	0.019	-0.002	≈ 1.000
Qwen2.5-VL	-0.025	-0.052	≈ 1.000
Qwen3-VL	**0.050**	**0.046**	≈ 1.000

Main finding

- **Qwen3-VL** is the strongest overall judge
- Parse rate is almost perfect for all judges

Prompt Variant Comparison

Prompt	Pearson rel.	Strict kappa rel.
prompt_v1	**0.026**	-0.0003
prompt_v2	0.004	-0.0062

Main finding

- **prompt_v1** is slightly better overall
- Prompt wording matters, but model choice matters more

Overall Interpretation

What the results suggest

- Proxy and VLM are **not random**, but their alignment is only moderate
- Easier categories:
  - complex
  - spatial
- Harder categories:
  - 3d_spatial
  - numeracy

Best-performing components

- Generator: **FLUX**
- VLM judge: **Qwen3-VL**
- Prompt variant: **prompt_v1**

Limitations and Future Work

Limitations

- Proxy is treated as reference ground truth, but it is not perfect
- Proxy errors may come from:
  - detection
  - segmentation
  - counting
- Agreement between proxy and VLM remains limited

Future work

- Compare against human annotation
- Improve the proxy pipeline
- Test stronger or more specialized VLM judges
- Improve prompting and calibration

Conclusion

Main conclusions

- Diffusion models still struggle with compositional reasoning
- 3d_spatial and numeracy are the hardest categories
- FLUX is slightly stronger overall under the proxy baseline
- Qwen3-VL is the best VLM judge
- prompt_v1 is the best prompt variant overall

Final message

- VLM-as-a-Judge is promising
- Proxy should be treated as an automatic baseline, not perfect truth
- Best evaluation uses both proxy-based checking and VLM-based semantic judgment

References

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